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# TOPIC 4

## Tectonic hazards

### Topics

[4.1 The structure of the Earth and the distribution of earthquakes and volcanoes](#)

[4.2 The processes and features associated with earthquakes and volcanoes](#)

[4.3 The impact of tectonic hazards](#)

[4.4 Managing the impacts of tectonic hazards](#)

This topic looks at:

- the structure of the Earth and the way in which earthquakes and volcanoes are formed
- the impacts and management of tectonic hazards.





# 4.1 The structure of the Earth and the distribution of earthquakes and volcanoes

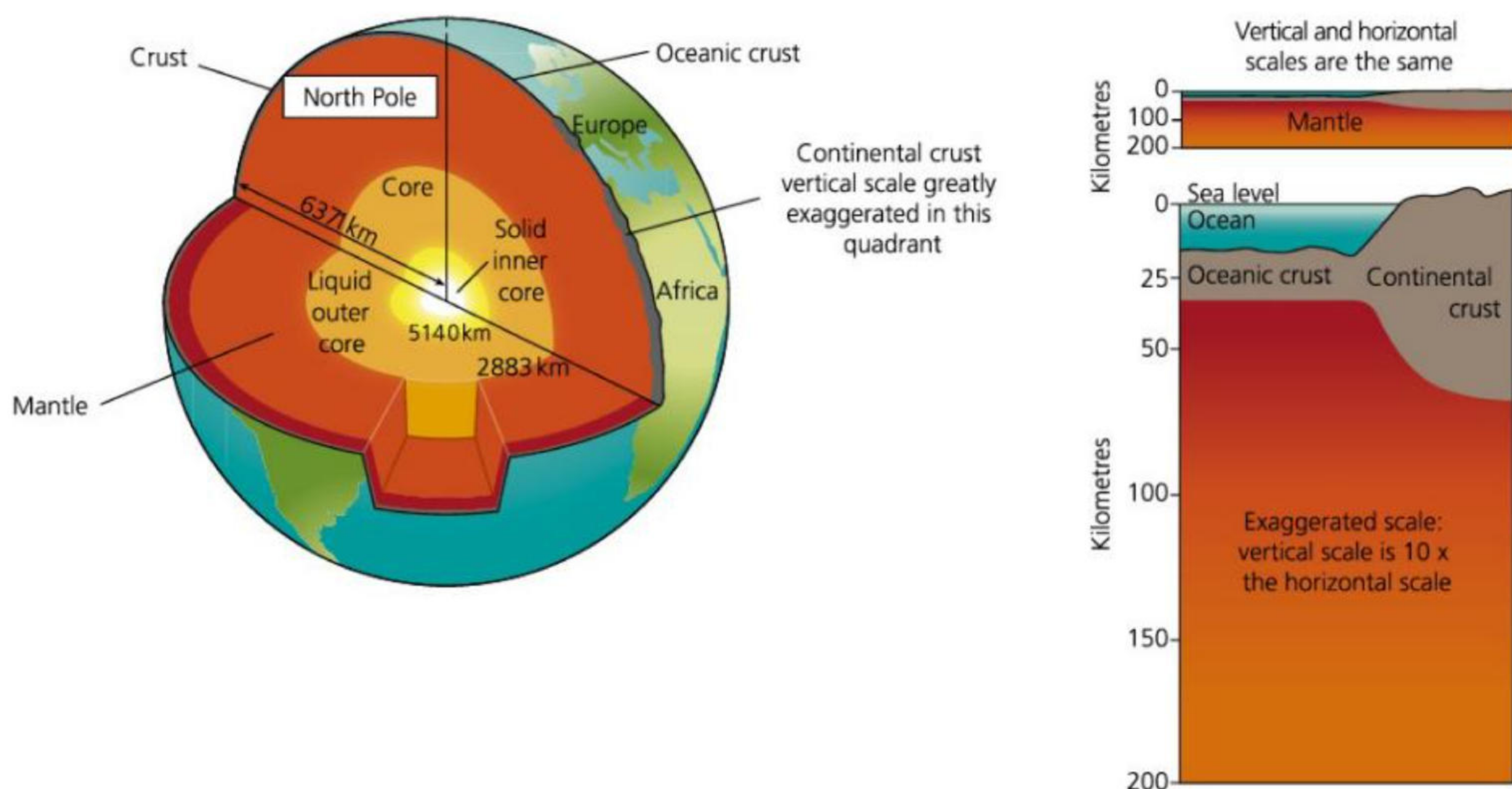
This chapter will explain:

- ★ the characteristics of the layers of the Earth, namely the inner core, outer core, mantle, crust and lithosphere
- ★ the main tectonic plates and how they move
- ★ the main types of plate boundary and how they relate to earthquakes and volcanoes.

## The characteristics of the layers of the Earth

There are four main layers within the Earth ([Figure 4.1](#)):

- » The core is divided into two parts:
  - The inner core is solid. It is five times denser than surface rocks.
  - The outer core is semi-molten.
- » The mantle is semi-molten and about 2900 km thick.
- » The crust is solid and its depth varies between 10 km and 70 km. It is divided into two main types:
  - Continental crust: Mostly formed of granite and less dense than the oceanic crust.
  - Oceanic crust: Mostly formed of basalt. As it is denser than the continental crust, it plunges beneath it when they come together.
- » The lithosphere is the solid outer part of the Earth and includes the brittle upper part of the mantle and the crust.



▲ **Figure 4.1** The structure of the Earth

## Activities

- 1 Compare the oceanic crust with the continental crust.
- 2 Compare the inner core with the outer core.

## Tectonic plates and their movement

[Figure 4.2](#) shows the location of tectonic plates and their direction of movement relative to one another. When the plates collide, they produce tectonic activity and new landforms. There are seven large plates (five of which carry continents) and a number of smaller plates:

- » The main plates are the Pacific, Indo-Australian, Antarctic, North American, South American, African and Eurasian Plates.
- » Smaller plates include the Caribbean, Iranian, Arabian and Juan de Fuca Plates.



Tectonic plates are constantly moving. Two main mechanisms for plate movement have been identified:

- » Convection currents
- » Gravitational sliding.

## Convection currents

Convection currents are partly responsible for the movement of tectonic plates. Heat produced by radioactive decay from the Earth's core causes rock (**magma**) to rise towards the crust and spread ([Figure 4.3](#)). The magma cools and sinks. This process is repeated in a cycle and is responsible for the movement of crustal plates.

## Gravitational sliding

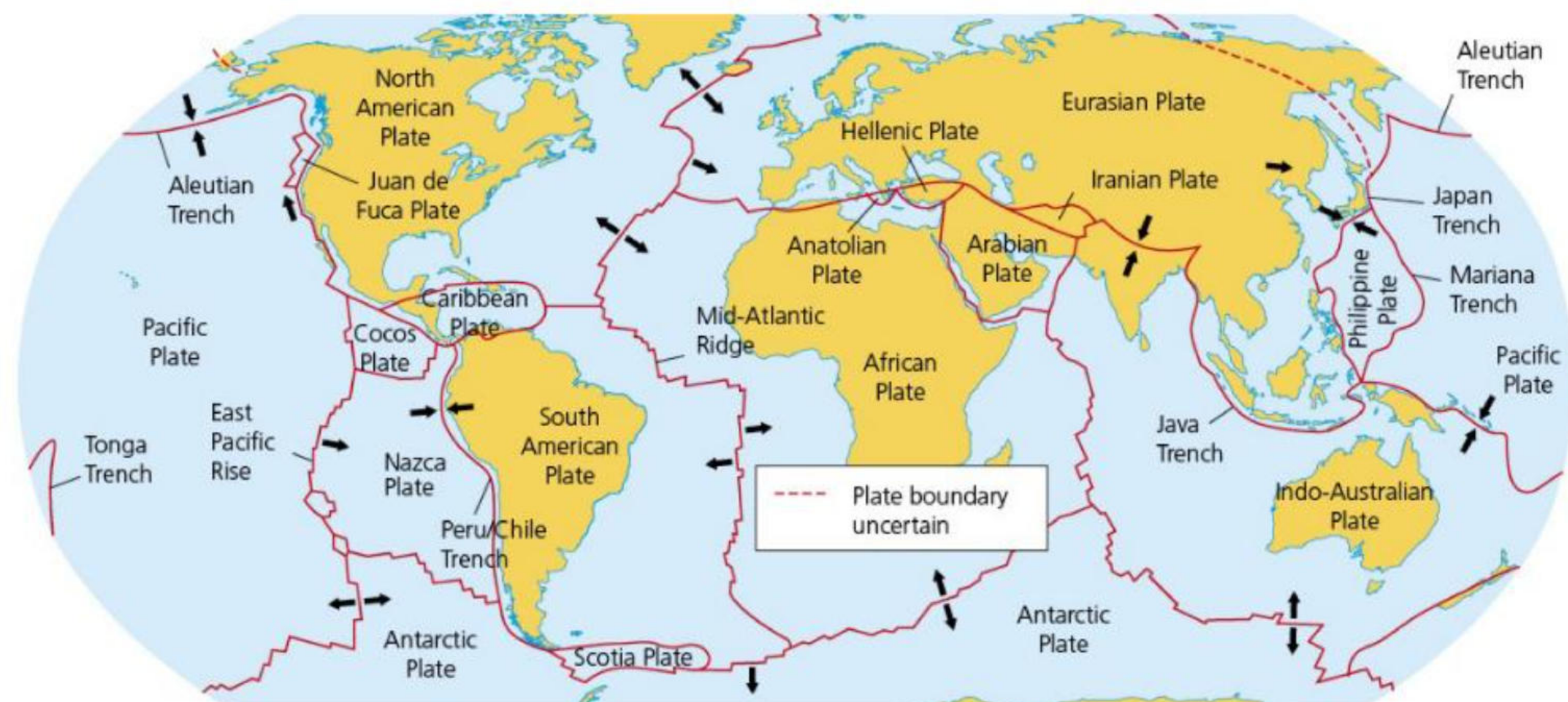
It is now believed that gravity is also important in driving tectonic plate movement. There are two ways in which gravitational sliding operates ([Figure 4.3](#)):

- » Ridge push: At mid-ocean ridges, new oceanic crust is pushed apart by rising magma from the mantle. As the mid-ocean ridge is higher than the ocean floor, gravity causes the plates to move downwards.
- » Slab pull: Gravity acting upon the colder and denser subducting (plunging) plate causes it to sink into the mantle and pull the remaining plate behind it.

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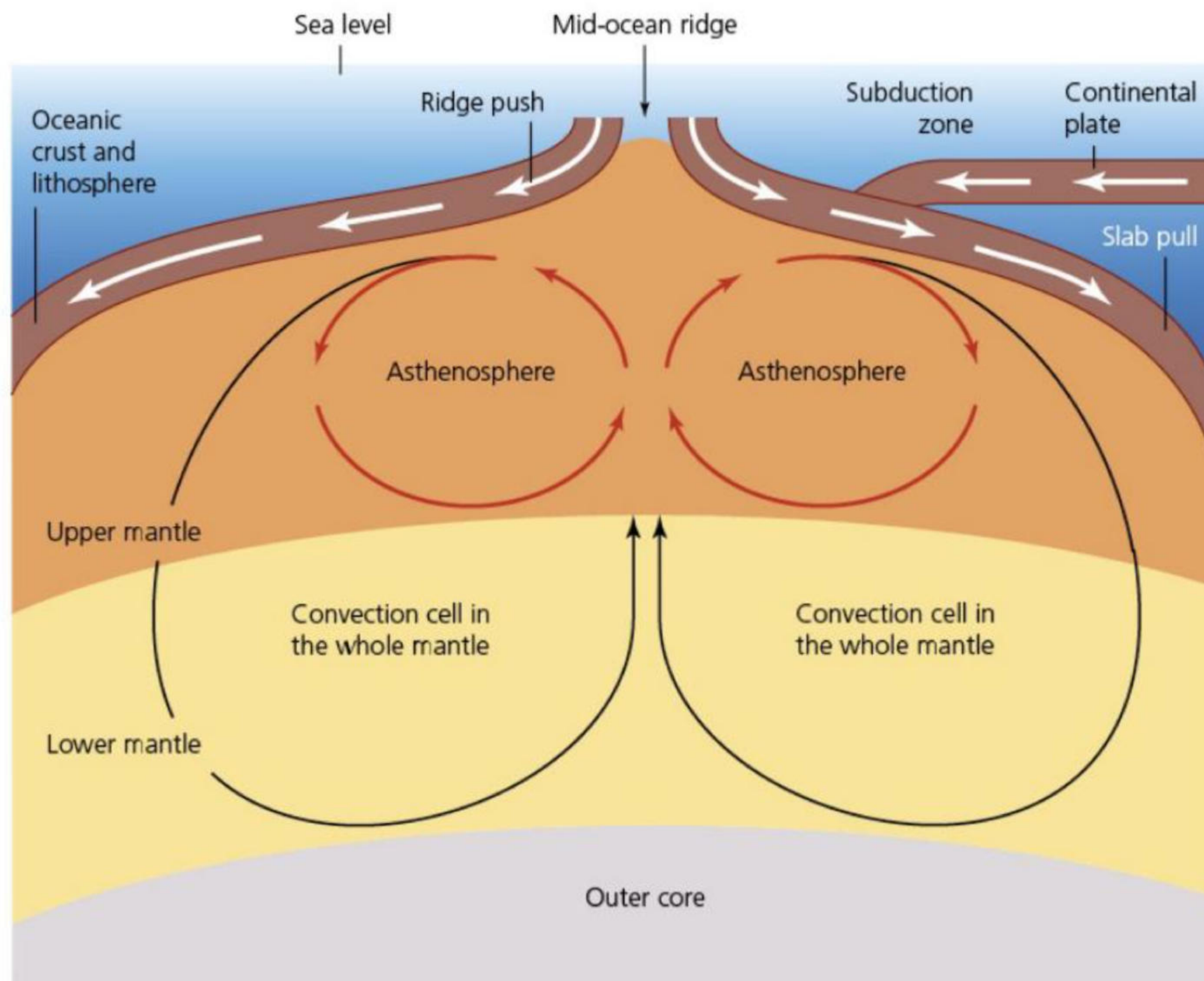
## Activities

- 1 Explain the difference between slab pull and ridge push.
  - 2 Identify one oceanic plate, one continental plate and one small plate.
- 



▲ **Figure 4.2** The world's main tectonic plates





▲ **Figure 4.3** Convection currents, ridge push and slab pull

## Types of plate boundaries

There are a number of different types of plate boundaries ([Table 4.1](#) and [Figure 4.4](#)). These include:

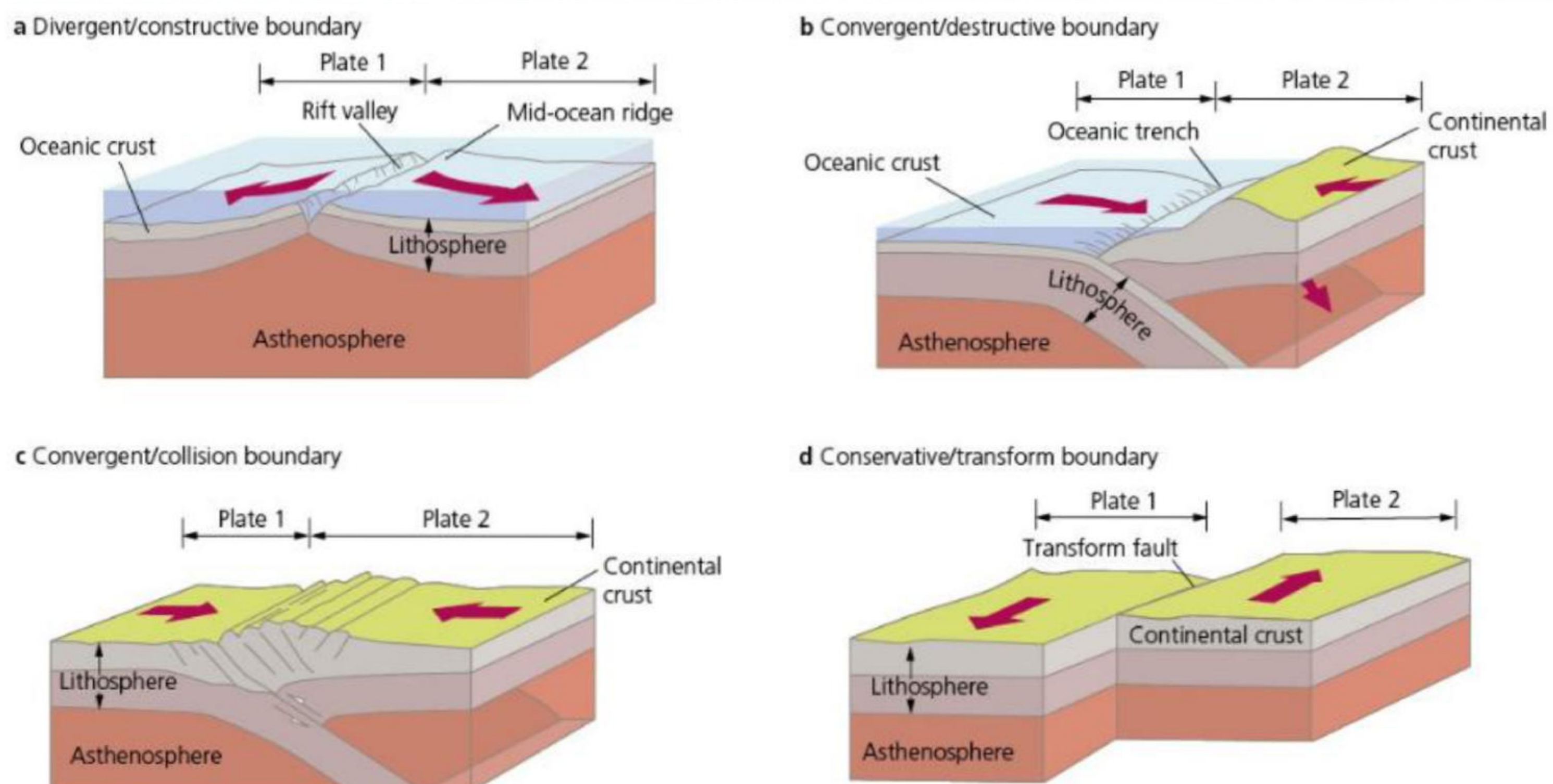
- » **divergent/constructive boundaries**, in which new oceanic crust is created
- » **convergent/destructive boundaries**, in which older crust is destroyed
- » **convergent/collision boundaries**, where plates are folded and crumpled
- » **conservative/transform boundaries**, where plates slip past each other, causing earthquakes to occur.

Different plate boundaries are associated with different tectonic activities: volcanic eruptions, folding and earthquake activity ([Figure 4.5](#)).

▼ **Table 4.1** The types of plate boundary

|                                        |                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Divergent/constructive boundary</b> | <ul style="list-style-type: none"> <li>● Two plates move apart from each other, causing sea floor spreading.</li> <li>● New oceanic crust is formed, creating mid-ocean ridges.</li> <li>● Volcanic activity is common, e.g. the Mid-Atlantic Ridge (Europe is moving away from North America).</li> </ul>                                                                                  |
| <b>Convergent/destructive boundary</b> | <ul style="list-style-type: none"> <li>● The oceanic crust moves towards the continental crust and sinks beneath it due to its greater density.</li> <li>● Deep-sea trenches and island arcs are formed.</li> <li>● The continental crust is folded into fold mountains.</li> <li>● Volcanic activity is common, e.g. the Nazca Plate is sinking under the South American Plate.</li> </ul> |
| <b>Convergent/collision boundary</b>   | <ul style="list-style-type: none"> <li>● Two continental crusts collide.</li> <li>● As neither can sink, they are folded up into fold mountains, e.g. the Indian Plate collided with the Eurasian Plate to form the Himalayas.</li> </ul>                                                                                                                                                   |
| <b>Conservative/transform boundary</b> | <ul style="list-style-type: none"> <li>● Two plates slip sideways past each other, but land is neither destroyed nor created, e.g. the San Andreas Fault in California.</li> </ul>                                                                                                                                                                                                          |





▲ **Figure 4.4** The types of plate boundary



▲ **Figure 4.5** Tectonic activities **a** A folded landscape, Himalaya foothills **b** Thingvellir Rift Valley, Iceland **c** Volcanic eruption of Soufrière, Montserrat, with the former capital city Plymouth in the foreground **d** Tourists standing by the boiling mud springs, Soufriere, St Lucia

## Activities

- 1 Name the type of plate boundary located:
  - a off the west coast of central America
  - b in the south Atlantic Ocean
  - c where the Indian Plate meets the Eurasian Plate.
- 2 Using [Figure 4.5 a-c](#), identify one location of:
  - a convergent/collision boundary
  - b a divergent/constructive plate boundary



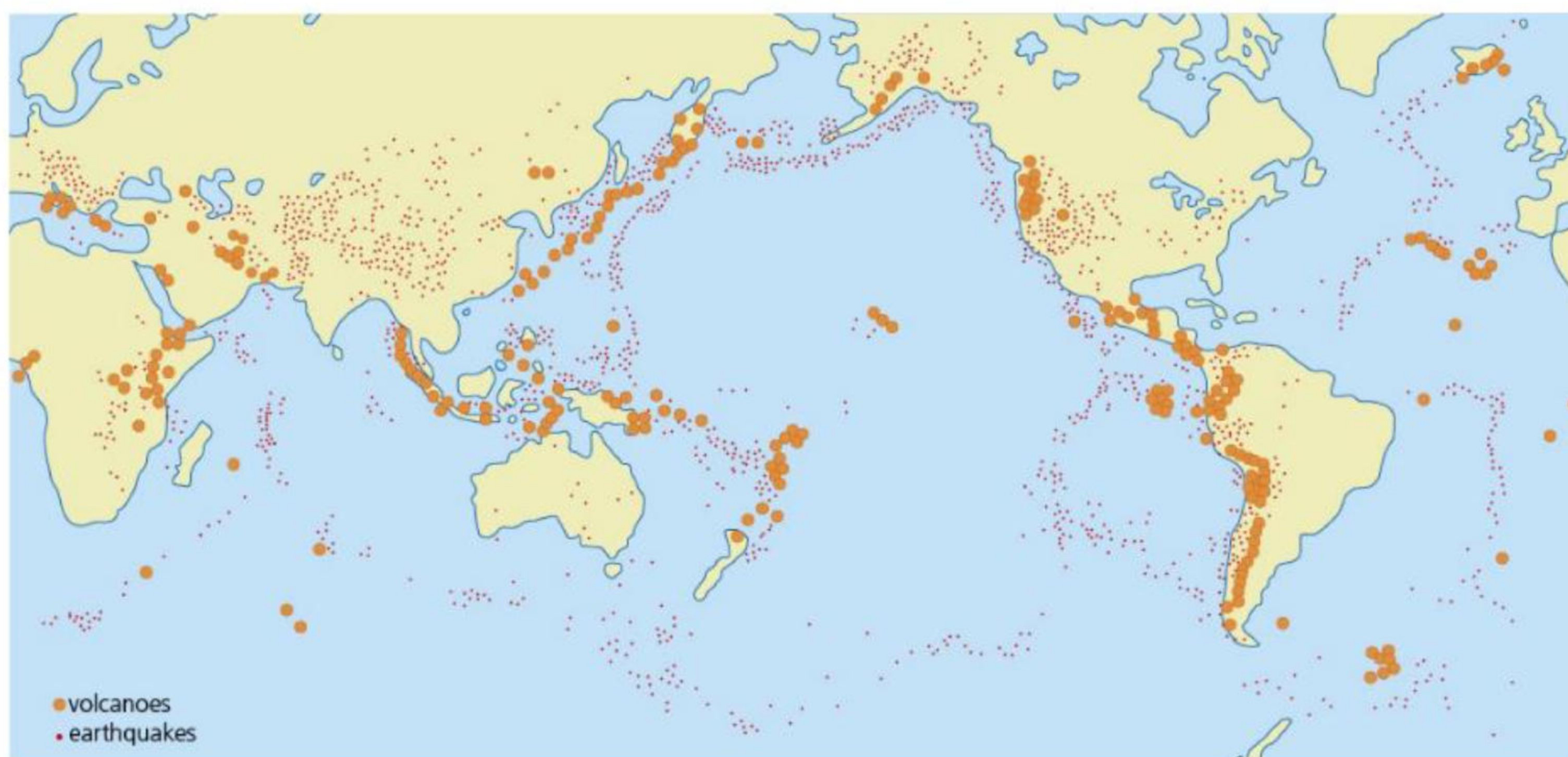
## The location of earthquakes

Earthquakes are located very unevenly around the world ([Figure 4.6](#)). About 500,000 earthquakes are detected each year by sensitive instruments. They are associated with all types of plate boundaries. Most of the world's earthquakes occur in linear chains along plate boundaries, such as along the west coast of South America. Some earthquakes appear in areas away from plate boundaries, such as in the Midwest of the USA. These earthquakes could still be related to plate movement, as the North American Plate is moving westwards.

Some earthquakes are the result of human activity. The building of large dams and deep reservoirs increases pressure on the ground. Mining removes underground rocks and minerals, which may cause collapse or subsidence of the overlying materials. Testing of nuclear weapons underground has been known to trigger earthquakes, too.

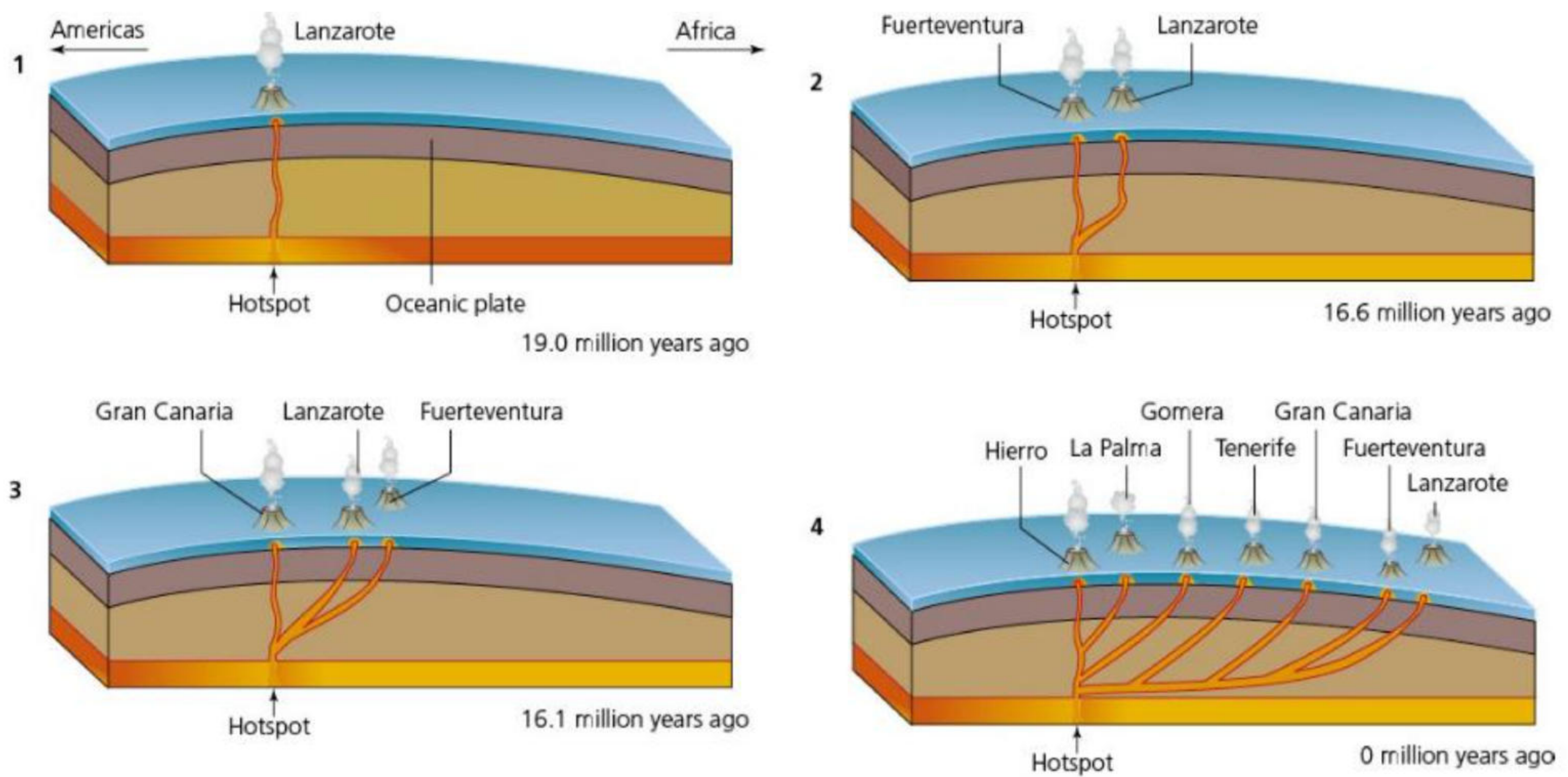
## The location of volcanoes

There are over 1300 active volcanoes in the world, many of them under the ocean (see [Figure 4.6](#)). Three-quarters of the world's active volcanoes are located in the 'Pacific Ring of Fire', the area around the Pacific Ocean. Good examples include Mount Pinatubo (Philippines), Krakatoa (Indonesia) and Popocatépetl (Mexico). These volcanoes are related to plate boundaries, notably convergent/destructive plate boundaries (for example Mount St. Helens in the USA and Soufrière in Montserrat in the Caribbean, see [Figure 4.8](#)) and divergent/constructive boundaries (for example Eldfell volcano on Heimaey, Iceland). The continuing eruption of Soufrière in Montserrat occurs at the boundary of the North American and Caribbean Plates. Some volcanoes, such as Mauna Loa and Kīlauea in Hawaii, and Teide on Tenerife, are located over hotspots. These are isolated plumes of rising magma that have burned through the crust to create active volcanoes ([Figure 4.7](#)).



▲ **Figure 4.6** World distribution of volcanoes and earthquakes





▲ **Figure 4.7** Volcanic activity associated with hotspots



▲ **Figure 4.8** The impacts of the Soufrière Hills volcano, Montserrat

## Activities

- 1 Describe the global distribution of earthquakes as shown on [Figure 4.6 \(page 91\)](#).
- 2 Explain why some volcanoes occur in locations away from plate boundaries.



## 4.2 The processes and features associated with earthquakes and volcanoes

This chapter will explain:

- ★ the processes at plate boundaries that cause earthquakes and volcanoes
- ★ the main characteristics of earthquakes
- ★ the main types of volcanoes
- ★ how volcanoes are classified
- ★ the main features of volcanoes
- ★ the hazards associated with volcanoes

### The processes at plate boundaries that cause earthquakes and volcanoes

At divergent/constructive plate boundaries two plates move apart from each other, causing **sea floor spreading**. New oceanic crust is formed, creating mid-ocean ridges and pushing the plates apart; volcanic activity is common. Less viscous, runny basaltic lava forms shield mountains. A narrow belt of shallow focus, low-magnitude earthquakes is formed. An example is the Mid-Atlantic Ridge (Europe is moving away from Northern America).

At convergent/destructive plate boundaries the oceanic crust moves towards the continental crust and is subducted (sinks) beneath it due to its greater density, pulling the rest of the plate behind it. Deep-sea trenches and island arcs are formed. Volcanic activity is common. More acidic, explosive lava results in cone volcanoes. **Pyroclastic flows** and **lahars** may occur ([Figure 4.9](#)). The plate movement compresses and folds the seafloor rocks/sediments, forming fold mountains. Earthquakes are a mix of shallow-, intermediate- and deep-focus and can be high magnitude. An example is the Nazca Plate sinking under the South American plate.

At a convergent/collision boundary two continental crusts collide. As neither can be subducted (sink), the crust and sediments become crumpled up into fold mountains. There is no volcanic activity due to the lack of magma. Earthquakes are shallow focus but powerful due to the friction as the plates move. A good example is the Indian Plate colliding with the Eurasian Plate to form the Himalayas. Prior to this, the Tethys Sea (continental crust) had collided into the Eurasian Plate. However, once it had fully subducted, the Indian Plate that it was dragging behind it collided with the Eurasian Plate to form a collision boundary.

At a conservative/transform plate boundary, two plates slip sideways past each other, but land is neither destroyed nor created. There is no volcanic activity due to the lack of magma. Earthquakes are shallow focus. Stress can build up close to the surface as plates 'get stuck' and this is released as earthquakes. The San Andreas Fault in California is a good example of a conservative/transform plate boundary, where the Pacific Plate is slipping past the North American Plate.



▲ **Figure 4.9** The impact of lahars at Plymouth, Montserrat

### Activities

- 1 State the types of plate boundaries where:



- a volcanic activity occurs
- b earthquakes occur.

2 Identify the type of location where volcanoes occur that is not a plate boundary.

## The main characteristics of earthquakes

Earthquakes involve sudden, violent shaking of the Earth's surface. They occur after a build-up of pressure causes rocks and other materials to give way. Most of this pressure occurs at plate boundaries, when one plate is moving against another. Earthquakes are associated with all types of plate boundary.

The **focus** refers to the place beneath the ground where the earthquake takes place. **Deep-focus earthquakes** are associated with **subduction zones**. **Shallow-focus earthquakes** are generally located along divergent/constructive boundaries and along conservative/transform boundaries ([Figure 4.10](#)). The **epicentre** is the point on the ground surface immediately above the focus.

Some earthquakes are caused by human activity, including nuclear testing, building large dams, drilling for oil/natural gas (fracking) and coal mining.

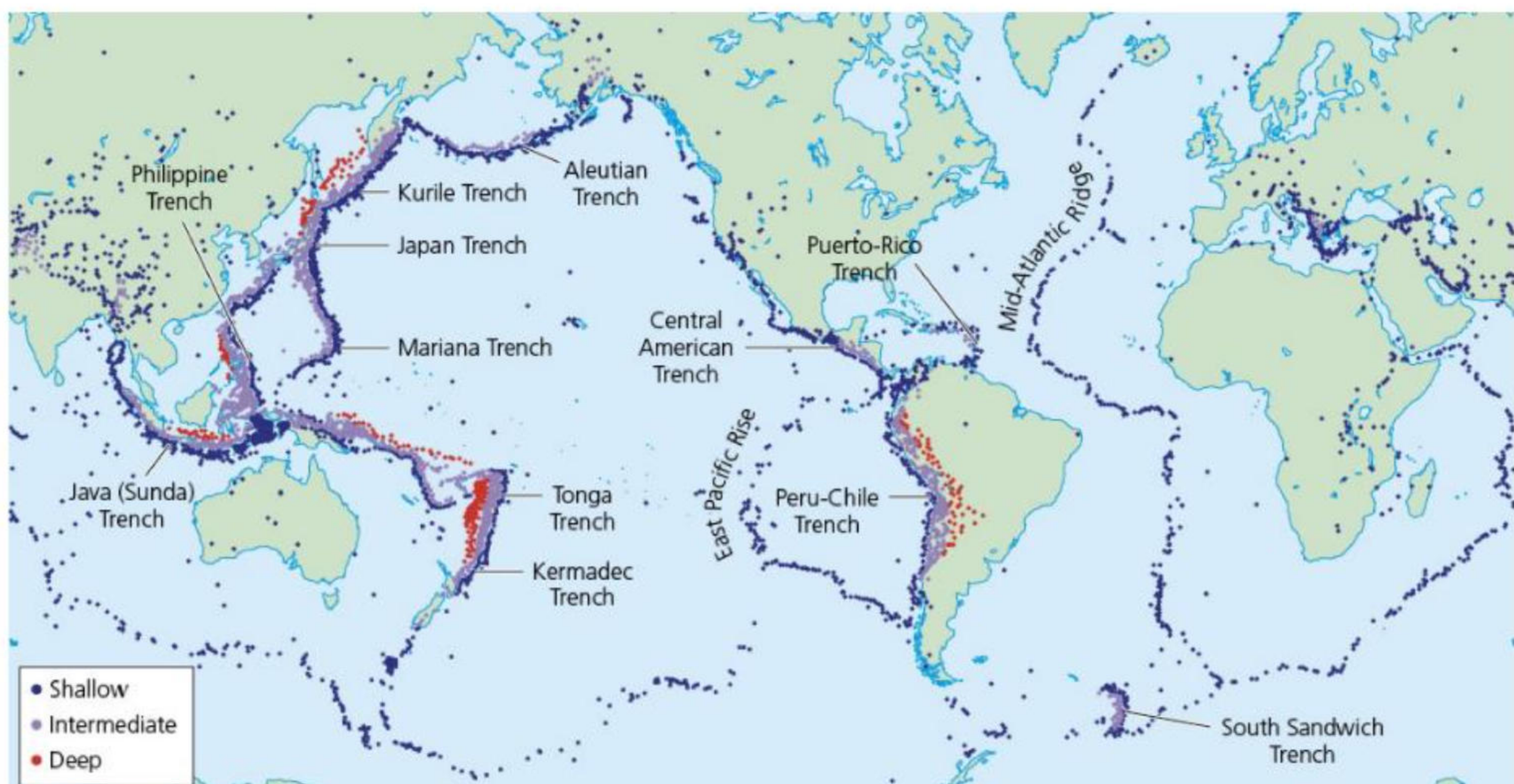
**Seismic waves** are the 'shock waves' generated by the release of energy following an earthquake. There are different types of seismic waves:

- » **Body waves** spread out from the focus of the earthquake.
- » **Surface waves** spread out from the epicentre. Surface waves are slower than body waves.

Body waves include **primary waves** (P-waves) and **secondary waves** (S-waves). Primary waves are compressional waves and are very fast (4–7 km/second). They can pass through liquids. In contrast, secondary waves are slower (2–5 km/second) and cannot pass through liquids.

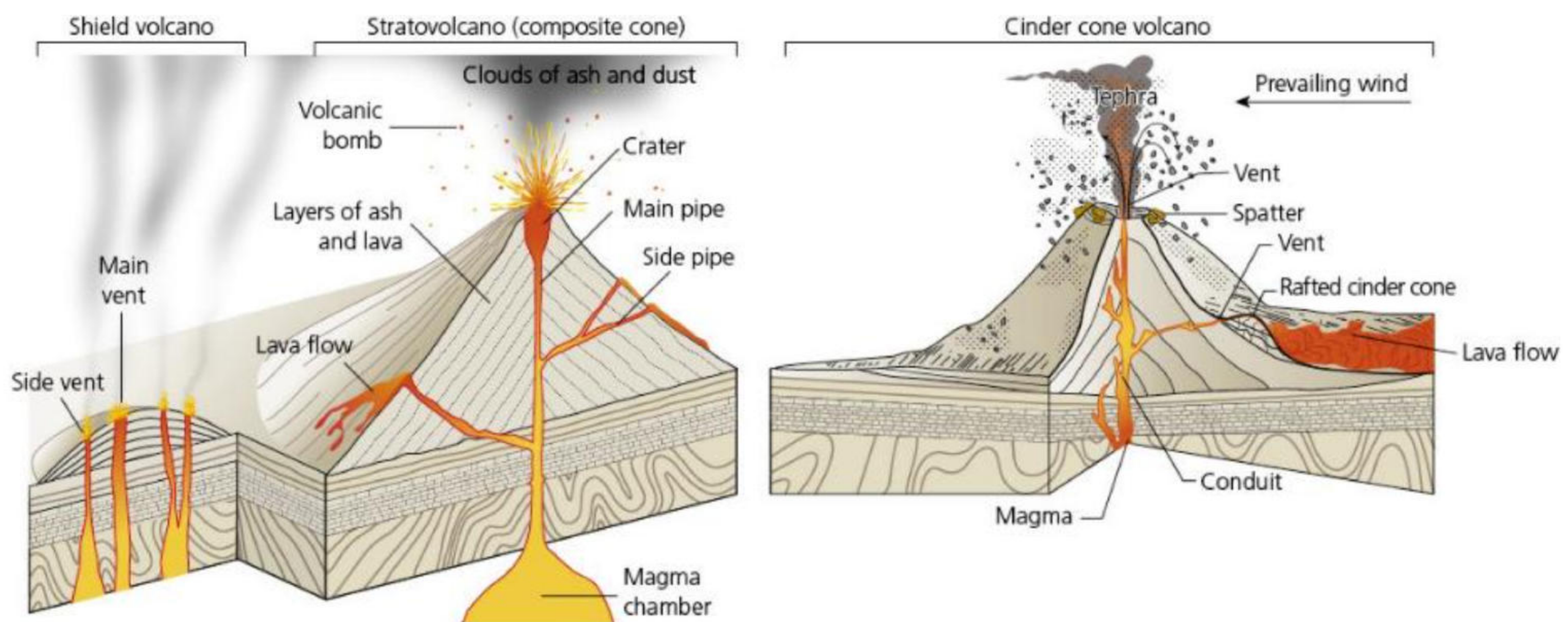
Surface waves are produced when the energy from an earthquake reaches the Earth's surface and causes a rolling or swaying motion. They are the slowest type of wave, but they cause the most damage.

Scientists use these waves to determine the exact location of the epicentre of the earthquake.



▲ **Figure 4.10** The distribution of shallow-, intermediate- and deep-focus earthquakes, 1900–2017





▲ **Figure 4.11** Shield, stratovolcano (composite cone) and cinder cone volcanoes

## Activities

- 1 Describe the distribution of shallow- and deep-focus earthquakes.
- 2 Distinguish between body waves and surface waves.

## The main types of volcanoes

The shape of a volcano depends on the type of lava it contains.

- » **Shield volcanoes** are gently sloping and produced by very hot, runny basaltic lava, for example Mauna Loa in Hawaii. They can cover a very wide area. They are made up of many layers of lava from repeated eruptions.
- » **Cinder cone volcanoes** are steep-sided, conical hills produced by thicker, less fluid material. They are made up of loose, pyroclastic fragments built up around a volcanic vent. Most cinder cones have a crater at the summit. Parícutin in Mexico was formed by ash and cinders building up into a symmetrical cone as a result of eruptions in the 1940s.
- » **Stratovolcanoes** (also known as **composite cone volcanoes**) are built up of many layers of hardened lava and tephra (volcanic ash, cinders, pumice and bombs), for example Mount Nyiragongo in Democratic Republic of the Congo and Mount Etna in Sicily. A stratovolcano is one that has had more than one eruption, and later eruptions may change the shape of the volcano. For example, Vesuvius was altered by the eruption in AD 79, forming a new version of itself, although some parts of the previous volcano survived the eruption. The shape of the volcano also depends on the amount of change there has been since the last volcanic eruption.

Cone volcanoes are generally associated with convergent/destructive plate boundaries, whereas shield volcanoes are characteristic of divergent/constructive boundaries and hotspots (areas of weakness in the middle of a plate).

## Activities

- 1 Distinguish between shield and cone volcanoes.
- 2 Identify the types of plate boundaries associated with shield and cone volcanoes.

## The classification of volcanoes

**Active volcanoes** are those that have erupted in recent times, such as Grindavik in Iceland, Kīlauea in Hawaii and Whakaari/White Island in New Zealand, which all erupted in 2023–24 and could erupt again.

**Dormant volcanoes** are volcanoes that have not erupted for many centuries but may erupt again, such as Mount Rainier in the USA.

**Extinct volcanoes** are not expected to erupt again. Kilimanjaro in Tanzania is an excellent example of an extinct volcano. The Le Puys region of France is an area of extinct volcanoes, which continue to influence settlements and tourism.

## The main features of volcanoes

A **crater** is the depression at the top of a volcano following a volcanic eruption. It may contain a lake. A **vent** is the channel that allows magma within the volcano to reach the surface in a volcanic eruption. Magma is the molten material from the Earth's interior. The **magma chamber** refers to the reservoir of



magma located deep inside the volcano. A **secondary cone** is a small volcano found on the side of the main volcano. For example, Mount Etna has several secondary cones ([Figure 4.12](#)).



▲ **Figure 4.12** Secondary cones on Mount Etna

## The hazards associated with volcanoes

**Lava flows** are streams of molten rock that come from an erupting vent. They can be erupted from an explosive or nonexplosive event. The speed at which lava flows move across the ground depends on many factors, such as the type of lava erupted, the steepness of the ground, whether the lava flows as a broad feature or in a confined channel, and the rate of lava production at the vent. In 2023–24, lava flows in Grindavik, Iceland ([Figure 4.13](#)) caused much disruption and led to the death of one person.



▲ **Figure 4.13** Lava flows, Grindavik, Iceland, 2023–24

**Tephra** refers to small fragments of rock ejected into the air by an erupting volcano. Most tephra falls back on to the slopes of the volcano, so enlarging it. But billions of smaller and lighter pieces less than 2 mm in diameter (ash) can be carried hundreds of kilometres by winds. **Ash falls** are potentially very hazardous and can disrupt human activities across thousands of square kilometres.

Ash fallout to the ground can cause significant disruption and damage to buildings, transportation, water and wastewater, power supply, communications equipment, agriculture and primary production. This can lead to potentially substantial societal impacts and costs, even at thicknesses of only a few millimetres. Additionally, when ingested, the fine-grained ash can cause health impacts to humans and animals.

Volcanic ash clouds are a major hazard to aviation. The 2010 eruption of Eyjafjallajökull in Iceland had a major impact on air travel. Over 300 airports in Europe were closed between 15 and 21 April 2010. More than 100,000 flights were cancelled, affecting around 7 million passengers and causing about US\$1.7 billion in lost revenue for the aviation industry.

A lahar is a fast-flowing mudflow made up of volcanic ash and water. The water may come from surface water (for example, from the lake in the volcano's crater), rain or melting snow. Lahars flow down valleys



and material is deposited over low-lying areas. Following the eruption of Mount Pinatubo in 1991, a tropical cyclone brought heavy rain and caused a number of lahars to form.



▲ **Figure 4.14** Pyroclastic flow on the Soufrière Hills volcano, Montserrat – a cone-shaped volcano formed by the subduction of the North American and South American Plates beneath the Caribbean Plate

Pyroclastic flows ([Figure 4.14](#)) are thick clouds of volcanic ash, pumice (volcanic rock) and gas that flow at high speed from a volcano into the valleys below. The temperature inside the pyroclastic flow can reach 700°C and flows can move at speeds of over 500 km/h. When they cool down, they form a layer of ash across the landscape that solidifies into pumice.

A **volcanic block** is a fragment of rock, over 64 mm in diameter, that is ejected in a solid condition from a volcano during an explosive eruption. Blocks are formed from material that was created during previous eruptions. In contrast, a volcanic bomb is a mass of partially molten rock (tephra) formed when a volcano ejects viscous fragments of lava during an eruption.

**Volcanic gases** may escape through vents in the ground. The most common volcanic gases include carbon dioxide, sulphur dioxide, hydrogen sulphide and carbon monoxide. Some of these gases are irritating or poisonous, or cause breathing problems. The release of sulphur dioxide may cause acid rain to form. The escape of carbon dioxide from Lake Nyos in Cameroon in 1986, possibly due to a small volcanic eruption, led to the death of over 1700 people and about 3500 livestock. This happened quickly so there was no warning and it was an infrequent event, but it was a high-magnitude event with a major impact. Fortunately, the gases did not spread to other areas.

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### Interesting notes

- ★ The greatest volcanic eruption in recorded history was Tambora in Indonesia in 1815. Some 50–80 km<sup>3</sup> of material was blasted into the atmosphere.
- ★ In 1883 the explosion of Krakatoa was heard from as far as 4776 km away.
- ★ The world's largest active volcano is Mauna Loa in Hawaii, which is 120 km long and over 100 km wide.

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### Activities

- 1 Distinguish between a volcano's crater and its chamber.
  - 2 Explain three hazards related to volcanic activity.
-



## 4.3 The impact of tectonic hazards

This chapter will explain:

- ★ the reasons why people live in areas at risk from earthquakes and volcanic eruptions
- ★ the impacts of earthquakes and volcanic eruptions
- ★ how the magnitude of tectonic events is measured.

### The reasons why people live in areas at risk from earthquakes and volcanic eruptions

All natural environments provide opportunities and challenges for human activities. Some countries, such as Iceland and the Philippines, were created by volcanic activity. People may choose to live in volcanic areas because they are useful:

- » Volcanic soils are rich, deep and fertile, and allow intensive agriculture to take place.
- » Volcanic areas are important for tourism.
- » Some volcanic areas are seen by people as being symbolic and are part of the national identity, such as Mount Fuji in Japan.

There are several ways of looking at people's vulnerability:

- » One view is that people choose to live in hazardous environments because they understand the environment. In this situation, people choose to live in an area because they feel there are potential advantages to living there and that these outweigh the risks. These advantages include:
  - fertile soils, which increase food production
  - geothermal energy and geothermal spas
  - minerals that can be used in the construction industry
  - volcanoes, which attract tourists
  - employment opportunities in tourism.
- » Another view is that some people live in hazardous environments because they have very little choice over where they live, as they cannot afford to move. Many communities have lived in these areas for generations.
- » Many people may not be aware of the potential risks associated with the hazard. The perception of the risk of the hazard may be low due to:
  - the infrequent occurrence of hazardous events
  - better monitoring, warnings, predictions and building design.

### Activity

Suggest contrasting reasons why some people live in volcanically active areas.

### The impacts of earthquakes

Hazards can be divided into two types – primary and secondary. Primary hazards are the direct hazards associated with natural events, while secondary hazards are the indirect hazards associated with natural events. [Table 4.2](#) shows the main hazards and impacts associated with earthquake activity.

▼ **Table 4.2** Hazards and impacts associated with earthquakes

| Primary hazards                                                                             | Secondary hazards                                                                                                                                                                                              | Impacts                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>● Ground shaking</li><li>● Surface faulting</li></ul> | <ul style="list-style-type: none"><li>● Ground failure and soil liquefaction</li><li>● Landslides and rockfalls</li><li>● Debris flows and mudflows</li><li>● Tsunamis (<a href="#">Figure 4.15</a>)</li></ul> | <ul style="list-style-type: none"><li>● Total or partial destruction of building structures</li><li>● Interruption of water supplies</li><li>● Breakage of sewage disposal systems</li><li>● Loss of public utilities such as electricity and gas</li><li>● Floods from collapsed dams</li><li>● Release of hazardous materials</li><li>● Fires</li><li>● Spread of chronic illness</li></ul> |





▲ **Figure 4.15** The 2004 tsunami caused widespread damage around the countries bordering the Indian Ocean

## Activity

Distinguish between primary and secondary hazards, giving an example of each.

## The impacts of volcanic eruptions

Likewise, there are several hazards related to volcanic activity ([Table 4.3](#)).

▼ **Table 4.3** The hazards and impacts associated with volcanic activity

| Primary hazards                                                                                                                                                                                                                                                                      | Secondary hazards                                                                                                                              | Impacts                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>● Pyroclastic flows</li> <li>● Volcanic bombs (projectiles)</li> <li>● Lava flows</li> <li>● Ash fallout</li> <li>● Volcanic gases</li> <li>● Lahars (mudflows) (see <a href="#">Figure 4.9 on page 93</a>)</li> <li>● Earthquakes</li> </ul> | <ul style="list-style-type: none"> <li>● Atmospheric ash fallout</li> <li>● Landslides</li> <li>● Tsunamis</li> <li>● Acid rainfall</li> </ul> | <ul style="list-style-type: none"> <li>● Destruction of settlements</li> <li>● Loss of life</li> <li>● Loss of farmland and forests</li> <li>● Destruction of infrastructure – roads, airstrips and port facilities</li> <li>● Disruption of communications</li> </ul> |

## Measuring the magnitude of tectonic events

The magnitude of an earthquake refers to its strength, and it remains unchanged with distance from the epicentre. The intensity describes the shaking caused by an earthquake at a given location, and this decreases with distance from the epicentre. The magnitude of earthquakes is measured using different scales.

### The Richter Scale

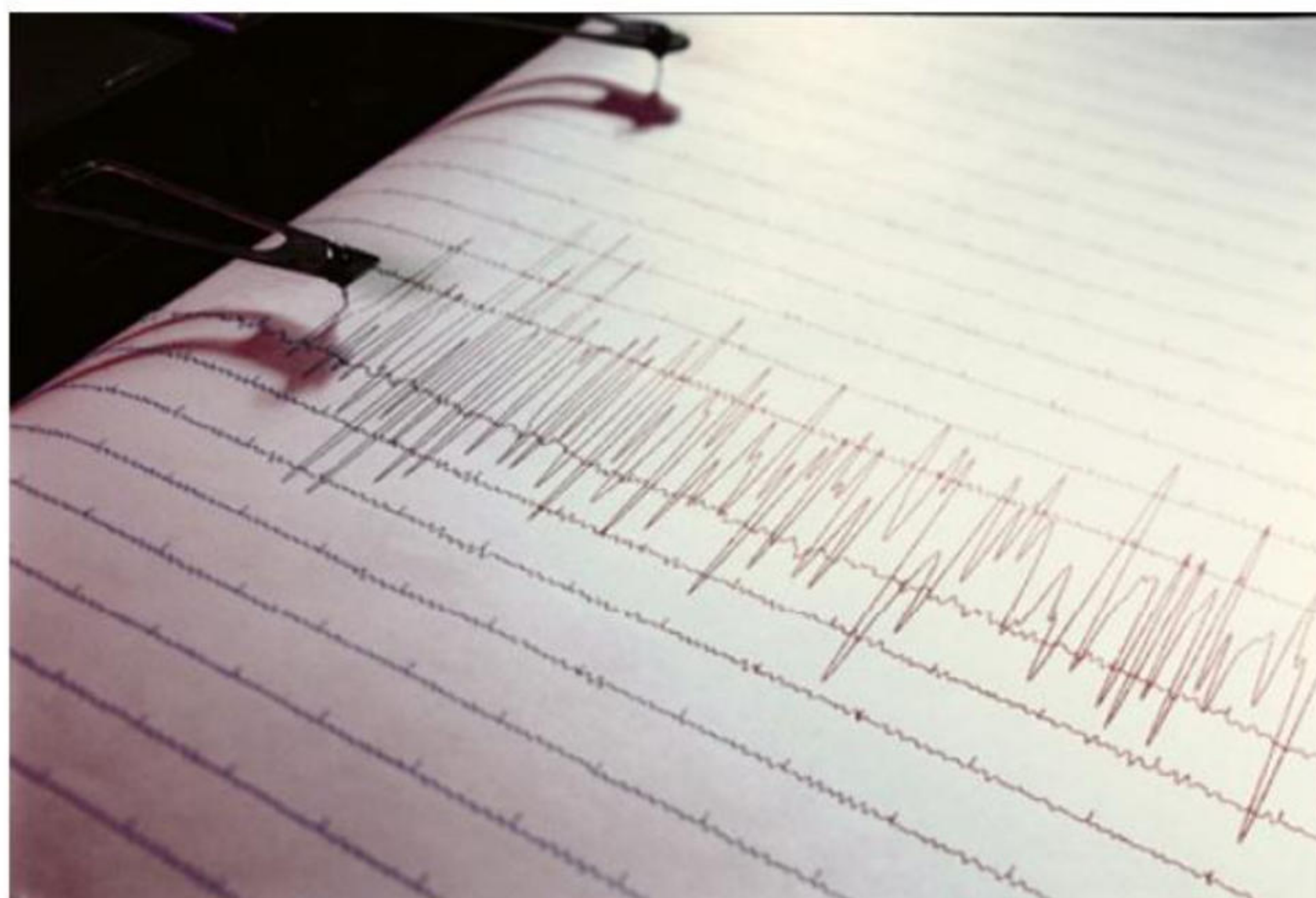
In 1935, Charles Richter of the California Institute of Technology developed the **Richter Scale** to measure the magnitude (strength or force) of earthquakes ([Table 4.4](#)). The strength or force of an earthquake is measured on a seismometer and shown on a seismograph ([Figure 4.16](#)). The Richter Scale is logarithmic. This means that an earthquake of 6.0 is ten times greater than one of 5.0, and 100 times greater than one of 4.0.

▼ **Table 4.4** The annual frequency of occurrence of earthquakes of different magnitudes, based on observations since 1900

| Descriptor | Magnitude (Richter Scale) | Average number/year | Hazard potential                            |
|------------|---------------------------|---------------------|---------------------------------------------|
| Great      | ≥8                        | 1                   | Total destruction, high loss of life        |
| Major      | 7.0–7.9                   | 18                  | Serious building damage, major loss of life |
| Strong     | 6.0–6.9                   | 120                 | Large losses, especially in urban areas     |



|                   |         |          |                                           |
|-------------------|---------|----------|-------------------------------------------|
| <b>Moderate</b>   | 5.0–5.9 | 800      | Significant losses in populated areas     |
| <b>Light</b>      | 4.0–4.9 | 6200     | Usually felt, some structural damage      |
| <b>Minor</b>      | 3.0–3.9 | 49,000   | Typically felt, but usually little damage |
| <b>Very minor</b> | <3.0    | 9000/day | Not felt, but recorded                    |



▲ **Figure 4.16** A seismograph reading from Soufrière Hills, Montserrat

## The Mercalli Scale

In contrast, the **Mercalli Scale** (Table 4.5) relates ground movement to things that you would notice happening around you. It is a subjective scale and lacks accuracy/reliability about the magnitude of the earthquake event. However, its main advantage is that it allows ordinary eyewitnesses to provide information on the strength of an earthquake. For example, a weak earthquake (II on the scale) might be felt by only a few people at rest in the upper floors of a building, whereas a violent earthquake (IX on the scale) is likely to cause considerable damage to buildings, with partial collapse.

▼ **Table 4.5** The Mercalli Scale

| Strength    | Observations                                                                                                                                                                                                                                      |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I</b>    | <ul style="list-style-type: none"> <li>● Rarely felt</li> </ul>                                                                                                                                                                                   |
| <b>II</b>   | <ul style="list-style-type: none"> <li>● Felt by people who are not moving, especially on the upper floors of buildings</li> <li>● Hanging objects may swing</li> </ul>                                                                           |
| <b>III</b>  | <ul style="list-style-type: none"> <li>● The effects are noticeable indoors, especially upstairs</li> <li>● The vibration is like that experienced when a truck passes</li> </ul>                                                                 |
| <b>IV</b>   | <ul style="list-style-type: none"> <li>● Many people feel it indoors; a few outside</li> <li>● Some are awakened at night</li> <li>● Crockery and doors are disturbed, and standing cars rock</li> </ul>                                          |
| <b>V</b>    | <ul style="list-style-type: none"> <li>● Felt by nearly everyone – most people are awakened</li> <li>● Some windows are broken, plaster becomes cracked and unstable objects topple</li> <li>● Trees may sway and pendulum clocks stop</li> </ul> |
| <b>VI</b>   | <ul style="list-style-type: none"> <li>● Felt by everyone – many are frightened</li> <li>● Some heavy furniture moves, plaster falls</li> <li>● Structural damage is usually quite slight</li> </ul>                                              |
| <b>VII</b>  | <ul style="list-style-type: none"> <li>● Everyone runs outdoors</li> <li>● Noticed by people driving cars</li> <li>● Poorly designed buildings are damaged</li> </ul>                                                                             |
| <b>VIII</b> | <ul style="list-style-type: none"> <li>● Damage to ordinary buildings – many collapse</li> </ul>                                                                                                                                                  |



|            |                                                                                                                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            | <ul style="list-style-type: none"> <li>● Well-designed buildings survive but suffer slight damage</li> <li>● Heavy furniture is overturned and chimneys fall</li> </ul>                                                     |
| <b>IX</b>  | <ul style="list-style-type: none"> <li>● Damage occurs even to buildings that have been well designed</li> <li>● Many buildings are moved from their foundations</li> <li>● Ground cracks and pipes break</li> </ul>        |
| <b>X</b>   | <ul style="list-style-type: none"> <li>● Most masonry structures are destroyed; wooden ones may survive</li> <li>● Railway tracks bend and water slops over banks</li> <li>● Landslides and sand movements occur</li> </ul> |
| <b>XI</b>  | <ul style="list-style-type: none"> <li>● No masonry structure remains standing; bridges are destroyed</li> <li>● Large cracks occur in the ground</li> </ul>                                                                |
| <b>XII</b> | <ul style="list-style-type: none"> <li>● Total damage</li> <li>● Waves are seen on the surface of the ground and objects are thrown into the air</li> </ul>                                                                 |

## The Moment Magnitude Scale

Scientists are increasingly using the **Moment Magnitude Scale** (M), which measures the amount of energy released in an earthquake. For every increase of 1 on the M scale, the amount of energy released increases by more than 32. Every increase of 0.2 represents a doubling of the energy released. Although this may be a more reliable and scientific measurement, a subjective scale may be enough for those who are affected by an earthquake, and the media still generally refer to the Richter Scale.

## The Volcanic Explosivity Index

The strength of a volcano is measured by the **Volcanic Explosivity Index** (VEI) ([Figure 4.17](#)). This is based on the amount of material ejected in the explosion, the height of the cloud it caused, and the amount of damage that results. Any explosion above level 5 is considered to be very large and violent.

A **supervolcano** is a volcano of VEI 8. The scale is logarithmic, so VEI 8 is ten times more powerful than VEI 7, 100 times more powerful than VEI 6, and 1000 times more powerful than VEI 5 (Mount Pinatubo, 1991). The last VEI 8 was 74,000 years ago (Mount Toba, Indonesia).

Supervolcanoes tend to be much larger than 'normal' volcanoes – the Yellowstone magma chamber, for example, is over 50 km wide. The likely impacts of a VEI 8 eruption include:

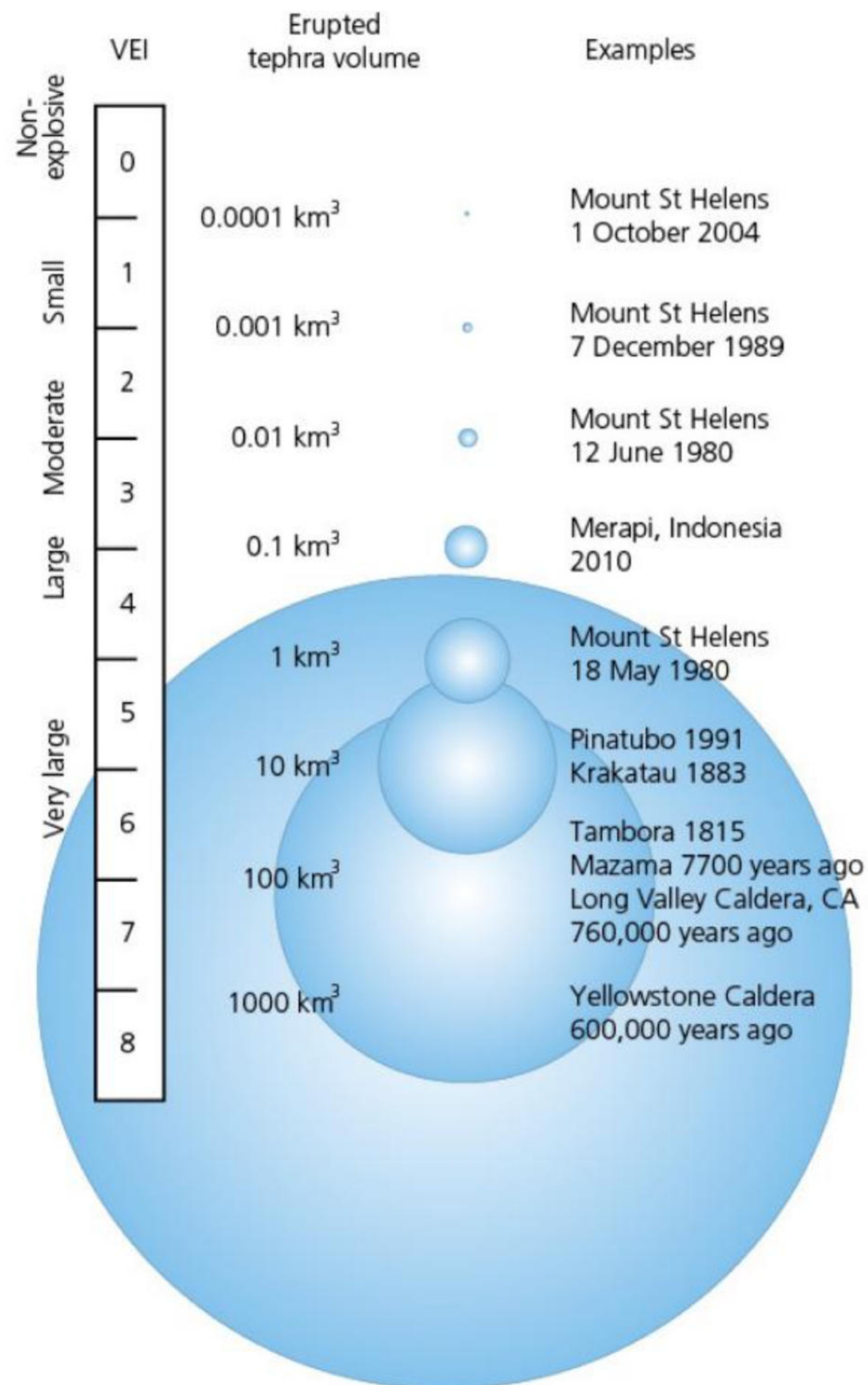
- » almost complete loss of life within about 1000 km of the eruption
- » destruction of all crops and livestock, leading to a global famine
- » economic and social devastation.

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## Activities

- 1 Distinguish between earthquake magnitude and intensity.
  - 2 The Richter Scale is logarithmic. How much stronger is an earthquake of 7.0 on the Richter Scale compared with one of 5.0?
-





▲ **Figure 4.17** The Volcanic Explosivity Index



## 4.4 Managing the impacts of tectonic hazards

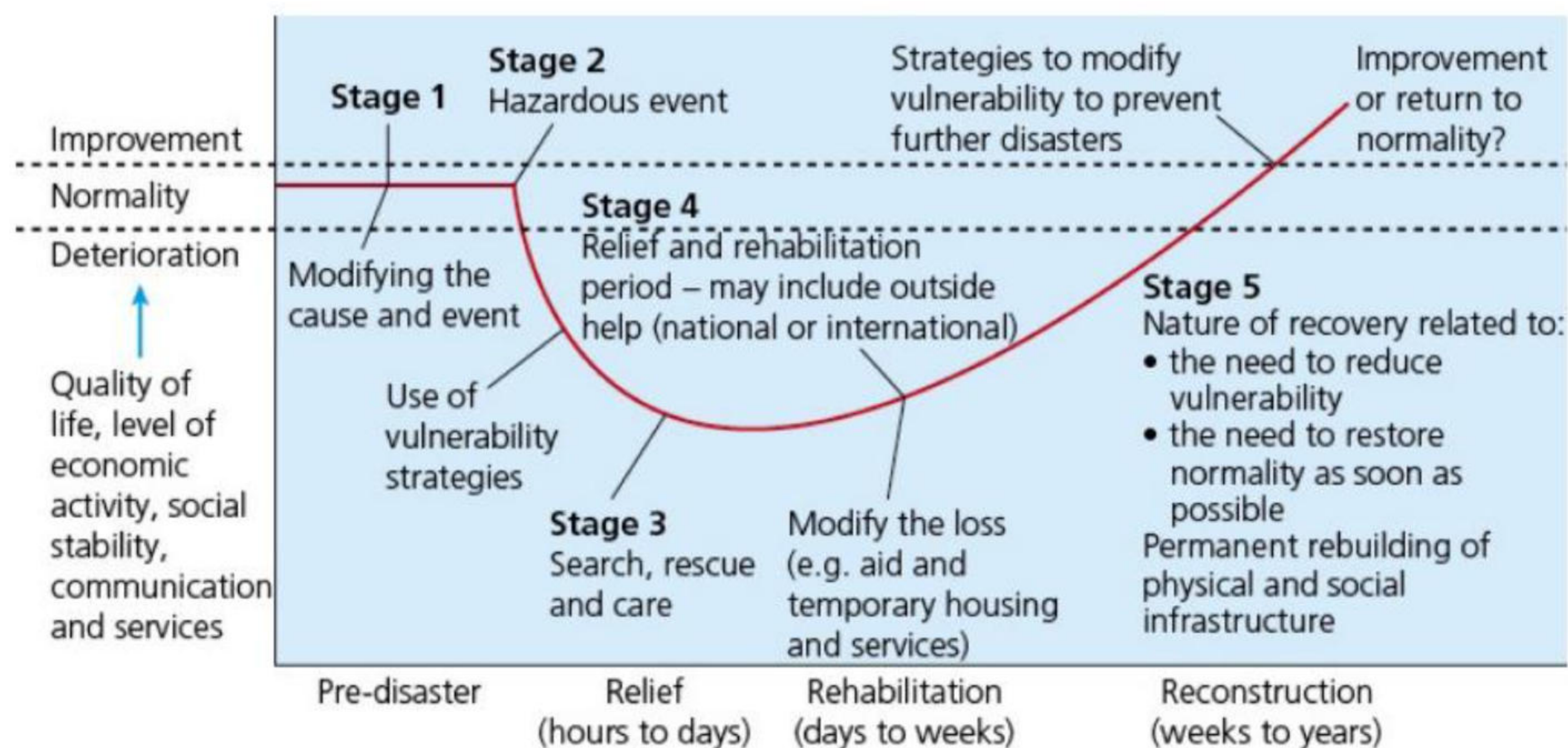
This chapter will explain:

- ★ the primary and secondary responses to tectonic hazards
- ★ the strategies and techniques used to manage the impacts of tectonic events
- ★ detailed specific example: Tiburon Peninsula earthquake, Haiti, 2021
- ★ detailed specific example: Mount Etna, Sicily, Italy.

### The primary and secondary responses to tectonic hazards

The Park Model shows the primary and secondary responses to natural hazards ([Figure 4.18](#)). The primary responses to tectonic hazards cover the short-term responses, that is search, rescue and relief, while the secondary responses refer to the long-term activities, such as rehabilitation and reconstruction.

- » Primary responses are those that occur in the first few days or weeks following a natural disaster. They include attempts to recover bodies, search and rescue, medical care for those with injuries, accommodation for those whose homes have been damaged or destroyed, and the provision of basic amenities such as food, clean water, shelter and healthcare.
- » Secondary responses usually occur after the first few weeks or months and can continue for years. They include the rebuilding of housing, education, and improving infrastructure. In many LICs, reconstruction may take decades.



▲ **Figure 4.18** The primary and secondary responses to hazards – the Park Model

### Activities

- 1 Distinguish between primary and secondary responses.
- 2 Identify whether the following are primary or secondary responses:
  - a Rebuilding homes
  - b Providing tents
  - c Providing toilets
  - d Rebuilding bridges.

### Managing the impacts of tectonic events

The impacts of volcanic eruptions can be reduced in several ways:

- » Spraying lava flows with water to cool them down, helping them to solidify. This was successfully carried out in Heimaey, Iceland. The islanders used sea water to spray on to the lava flow and managed to divert it away from reaching the town. In contrast, some residents of Goma in the Democratic Republic of the Congo tried to spray water on to lava flows from the Nyiragongo volcano, but this proved unsuccessful as they had neither enough water nor the infrastructure to deal with the advancing lava flows.
- » Digging diversion channels to divert lava flows away from settlements: This has been used successfully



on Mount Etna, Sicily, although the lava needs to be diverted away from human habitation and infrastructure.

- » Adding 'cold' boulders to a lava flow in an attempt to cool the lava and stop it moving.

However, if the eruption is a pyroclastic flow, there is little that can be done to prevent the impacts other than evacuation.

## Predicting volcanoes

The main methods of predicting volcanoes include:

- » using seismometers to record swarms of tiny earthquakes that occur as the magma rises
- » using chemical sensors to measure increased sulphur levels
- » using lasers to detect the physical swelling of the volcano
- » measuring small-scale uplift or subsidence, changes in rock stress and changes in radon gas concentration
- » using ultrasound to monitor low-frequency waves in the magma resulting from the surge of gas and molten rock, as happened with Mount Pinatubo, El Chichón and Mount St. Helens.

Overall, relatively few people are killed by volcanic eruptions. This may be because most of the world's active volcanoes are monitored, especially in areas with high population density. Nevertheless, volcanoes can erupt without warning. If Vesuvius in Italy were to erupt again, millions of people could be affected. In addition, high-magnitude events, such as a VEI 8 supervolcano, will occur again at some point, and there is little that can be done to reduce some of the global impacts of such an event.

---

## Activities

- 1 Outline two ways of controlling lava flows.
  - 2 Outline two ways in which volcanoes can be predicted.
- 

## Preparing for earthquakes

The main ways of preparing for and managing earthquakes include:

- » better forecasting, warning, evacuation and emergency procedures
- » improving building design and safe houses
- » land-use planning for building location.

### Forecasting and warning

There are a number of ways of predicting and monitoring earthquakes. These include:

- » measuring crustal movement – the small-scale movement of plates
- » recording changes in electrical conductivity
- » noting strange and unusual animal behaviour, for example among fish (such as carp) or toads
- » checking historical evidence – there are possibly trends in the timing of earthquakes in some regions.

### Evacuation and emergency procedures

Loss of life can be minimised by ensuring that the population is educated in emergency evacuation procedures. For example, schools in Japan practise earthquake drills, and there is disaster education in many schools, colleges and states in the USA.

The US Centers for Disease Control and Prevention (CDC) provides information on preparing for hazards, for example 'drop, cover, and hold on' in the case of an earthquake. In an earthquake, people should get outside if possible. If trapped indoors, they should shelter under a sturdy object such as a kitchen table. When the shaking has stopped, they should get outside and move away from the building.

### Building design

Buildings can be designed to cope with the shock waves that occur in an earthquake. For example, single-storey buildings are more able to absorb shock waves than multi-storey buildings as the potential for swaying is reduced. Also, tall buildings can be built with a 'soft storey' at the bottom, such as a car park on raised pillars. This soft storey may collapse in an earthquake so that the upper floors sink down on to it, which cushions the impact.

Building reinforcement strategies include building on foundations that are laid deep into the underlying bedrock, and steel frames that can withstand shaking can also be used.

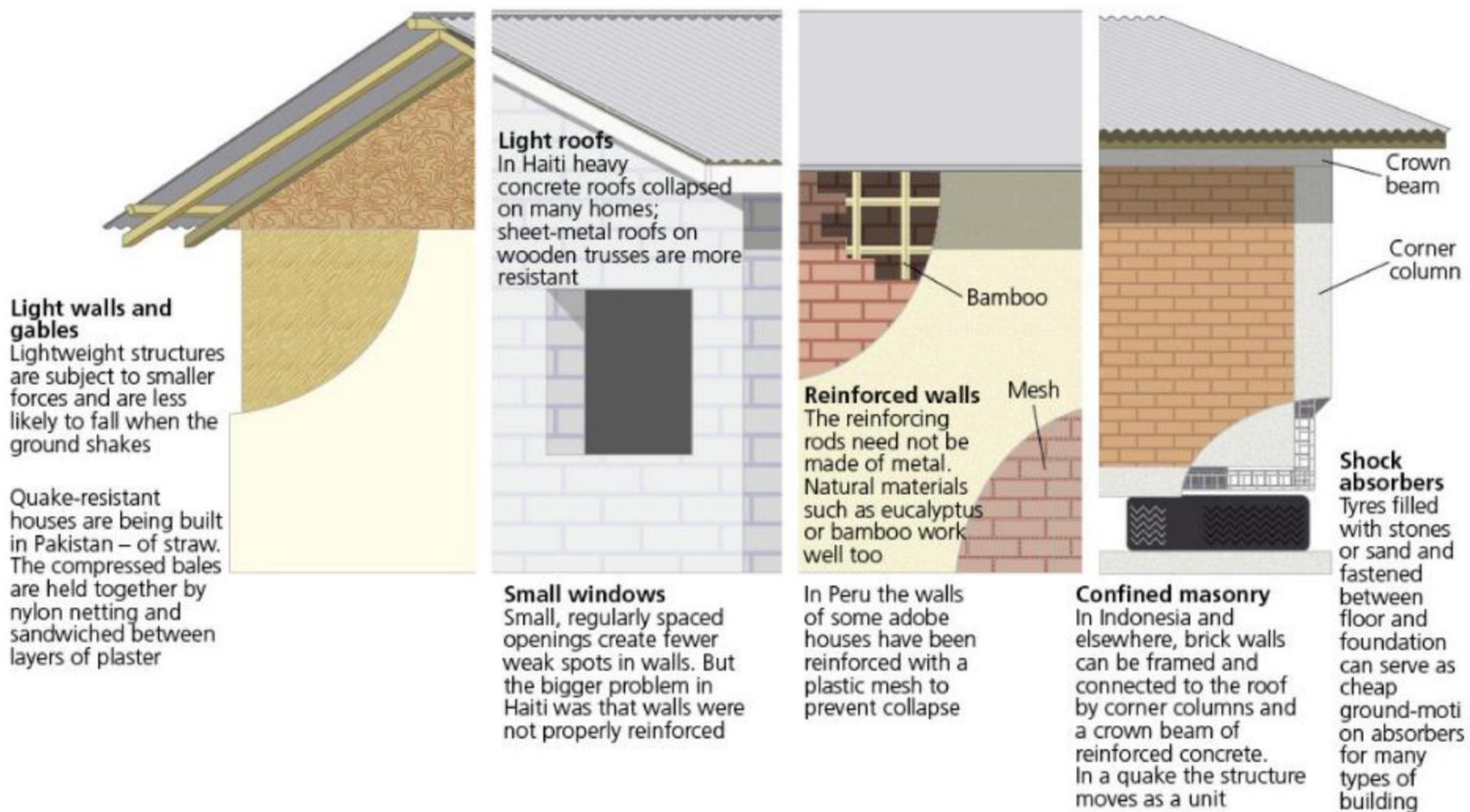
In an earthquake, high-rise buildings respond slowly, and shock waves are increased as they move up the building. If buildings are too close together, vibrations may be amplified between buildings and increase the damage. Therefore maximising space between buildings can help minimise this amplification.

### Safe houses

Poverty, corruption and poor governance mean that buildings in many countries are not earthquake-proof. Billions of people live in houses that cannot withstand shaking. Yet safer ones can be built cheaply, using straw, adobe (cheap, sun-dried brick) or old tyres, by applying a few general principles ([Figure 4.19](#)).



In rich cities in fault zones, the added expense of making buildings earthquake-resistant has become a fact of life. Concrete walls are reinforced with steel, for instance, and a few buildings even rest on elaborate shock absorbers. Strict building codes were credited with saving thousands of lives when a magnitude 8.8 earthquake hit Chile in February 2010. But in less developed countries, like Haiti, conventional earthquake engineering is often unaffordable, even though there are some cheap solutions.



▲ **Figure 4.19** Safe-house design

In Peru in 1970 an earthquake killed more than 70,000 people, many of whom died when their houses crumbled around them. Heavy, brittle walls of traditional adobe cracked instantly when the ground started to move. Existing adobe walls can be reinforced with a strong plastic mesh installed under plaster. During an earthquake, these walls crack but do not collapse, allowing occupants to escape. Plastic mesh could also work as a reinforcement for concrete walls in Haiti and elsewhere.

Researchers in India have successfully tested a concrete house reinforced with bamboo. A model house for Indonesia rests on ground-motion dampers – old tyres filled with bags of sand. Such a house might be only a third as strong as one built on more sophisticated shock absorbers, but it would also cost much less – and so is more likely to get built in Indonesia. In northern Pakistan, straw is available. Traditional houses are built of stone and mud, but straw is far more resilient, and warmer in winter.

### Building location

Land-use planning is another way of reducing earthquake risk. Densely populated areas and important services, such as hospitals and fire services, should not be built close to known fault lines or areas prone to landslides. Areas with weak rocks, faulted (broken) rock and soils should, ideally, be avoided for buildings. In California, since 1972, no building can be built within 17 m of a surface fault line, although this would not offer much protection from an earthquake.

Ultimately, although it is possible to suggest areas where most earthquakes are likely to occur, it is not possible to predict with certainty exactly where they will occur, when they will occur or their magnitude. However, taking precautions may help reduce their impact.

Certain areas are very much at risk from earthquake damage – notably areas with weak rocks, faulted rocks and soft soils. Many oil and water pipelines in tectonically active areas are built on rollers so that they can move with an earthquake rather than fracture ([Figure 4.20](#)).

## Activities

- 1 Identify ways in which earthquakes can be predicted.
- 2 Explain the term 'safe house'. Briefly explain how houses can be made 'safe'.





▲ **Figure 4.20** A pipeline on rollers to allow movement during an earthquake

### Detailed specific example

#### Tiburon Peninsula earthquake, Haiti, 2021

Haiti is located on the Gonâve Microplate, a small strip of the Earth's crust that is squeezed between the North American and Caribbean Plates. This makes it vulnerable to earthquakes. In August 2021, a 7.2 magnitude earthquake struck the country's Tiburon Peninsula. It was a shallow-focus earthquake (10 km deep) and the epicentre was about 150 km west of the capital Port-au-Prince ([Figure 4.21](#)). The earthquake was caused by the eastward movement of the Caribbean plate relative to the North American plate.

The earthquake caused around 2250 deaths, more than 12,750 injuries and damage estimated at US\$1.5–1.7 billion. There were over 900 aftershocks, two of which were in 2022, killing a further two people. The earthquake was followed by the impact of Hurricane Grace the next day, which triggered thousands of landslides. At least 137,500 buildings were damaged or destroyed. Les Cayes, Haiti's third-largest city, was badly affected, with over 37,000 homes destroyed and 46,000 others damaged, see [Figure 4.22](#). More than 50 medical facilities were damaged and six destroyed, and over 1000 schools were damaged or destroyed.

Shelter was the greatest long-term need. Emergency services were required given the needs of the population – including food, water, medicines and Covid-19 vaccines. Rebuilding is ongoing, including to infrastructure (communications, transport, WASH (water, sanitation and hygiene) and electrical supplies).

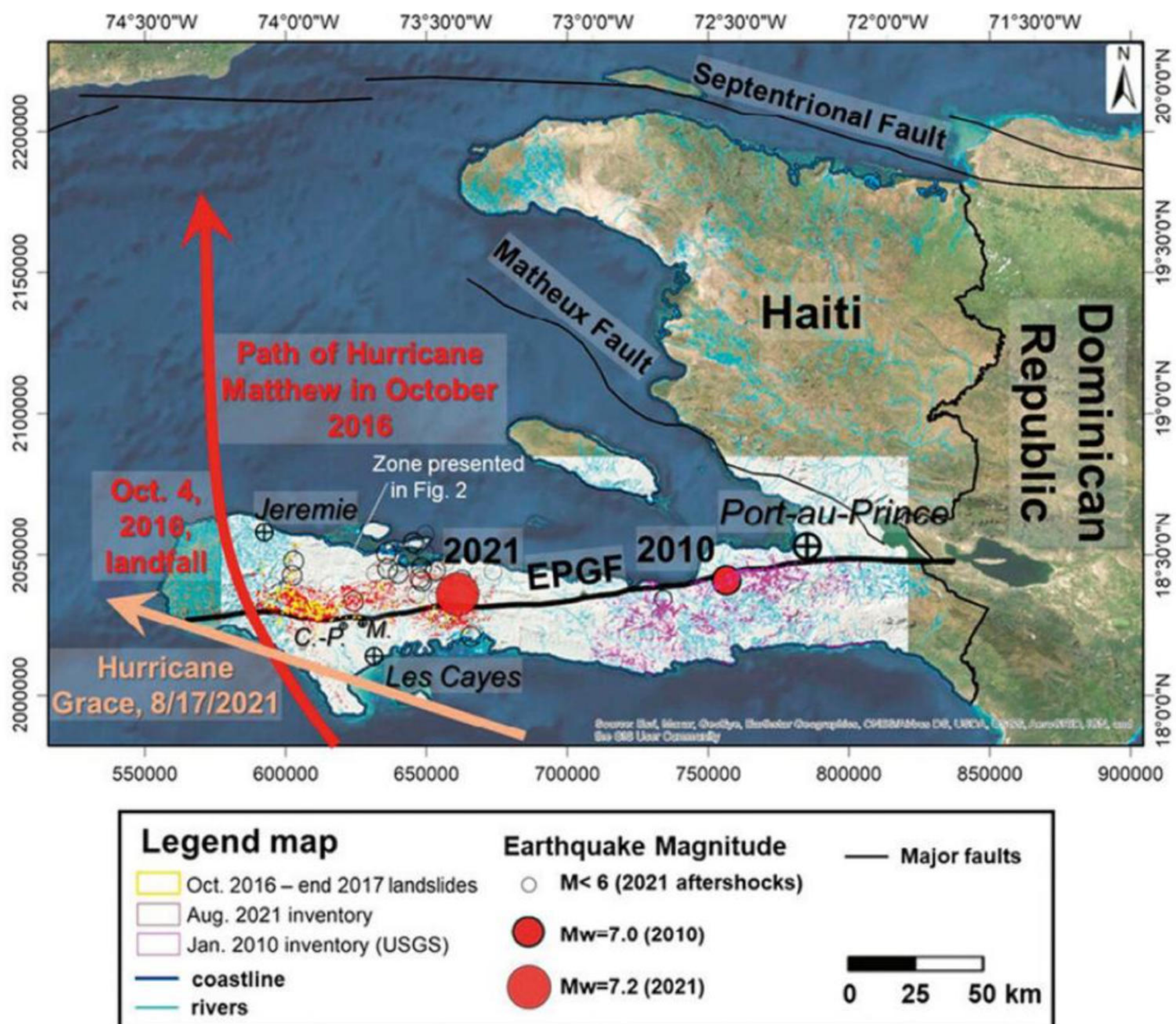
Six months after the earthquake, Haiti began to look at long-term recovery and reconstruction. It needed around US\$2 billion to rebuild social services and housing, health and education infrastructure, and food supply. After that, aid funds were spent on agriculture, industry and key infrastructure. Haiti's financial situation meant it was reliant on international aid for its recovery. It is the lowest-income country in the Western Hemisphere and is included by the UN on its list of the 45 least developed countries in the world, the only country from the Caribbean.

However, many donor countries were facing their own financial problems or were less able or unwilling to provide aid. In addition, Haiti was competing with other countries for funding, notably Afghanistan, Ethiopia, Ukraine, and Turkey and Syria (following the earthquake in 2023).

Research using remote sensing of the damage to buildings found that those with deep foundations and timber roofs fared best during the earthquake. The research, undertaken by international scientists with the aid of local volunteers who took notes and uploaded photographs, has helped ensure that new buildings will be more earthquake-proof for the future.

According to the World Bank, the 2021 Disaster Risk Management and Reconstruction (DRM) Project for Haiti focused on supporting Haiti to improve its disaster-response capacity, and enhancing the resilience of critical transport infrastructure.





▲ **Figure 4.21** The location of the Tiburon Peninsula earthquake and the track of Hurricane Grace

Key activities and systems developed under the project included risk assessments, communications and early warning systems, and improved road sections. In response to the August 2021 earthquake, the government was able to:

- mobilise Municipal Civil Protection Committees
- conduct a rapid and sound assessment of damages and losses
- achieve greater coordination among groups and individuals working in the DRM system.

## Activities

- 1 Explain why earthquakes occur in Haiti.
- 2 Suggest how government actions after the 2021 earthquake helped manage the recovery process.





▲ **Figure 4.22** Search and rescue can be physically demanding as well as extremely dangerous

### Mount Etna, Sicily, Italy

Mount Etna is an active stratovolcano on the east coast of the island of Sicily, Italy. It is located on the convergent plate margin of the African and Eurasian Plates (Figure 4.23). It is one of the world's most active volcanoes. In 2021 it increased in height by approximately 30 m, to around 3369 m. In historic times, only 77 deaths have been caused by its eruptions, the last two being in 1987. Nevertheless, Etna is a very active volcano.

There are different types of eruption at Etna. Most occur at one of the five craters near the summit (Figure 4.24). There are about 300 side vents, ranging in size from large craters to small holes. Ironically, the highly explosive summit eruptions are less hazardous than the flank eruptions, as these can occur close to inhabited areas. Since 2000 there have been at least four periods of flank eruptions and six periods of summit eruptions.

Volcanic activity has occurred at Etna for around 500,000 years. It evolved in three main phases from a shield volcano caused by submarine fissure eruptions to a stratovolcano caused by explosive activity. Over time, volcanic activity has moved from the southeast to the northwest.



▲ **Figure 4.23** The location of Mount Etna and tectonic plates





▲ **Figure 4.24** The craters near the summit of Etna

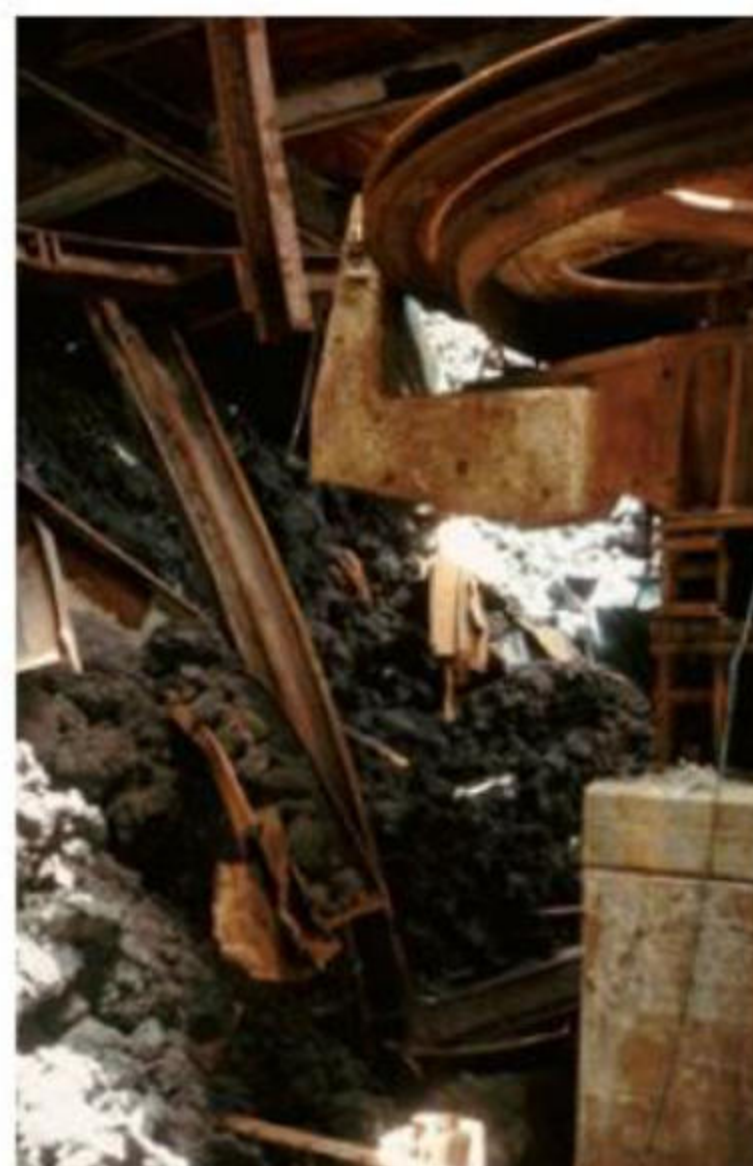
Around 8000 years ago, a massive landslide caused a large depression on the side of Etna – the Valley of the Ox – and triggered a large tsunami. Major eruptions have caused the summit to collapse many times, forming calderas. A collapse 2000 years ago formed the Piano Caldera, although this has largely been filled in by subsequent lava flows.

In 1971 lava flows buried the Etna Observatory and destroyed the Etna cable car. The 1991–93 eruptions threatened the town of Zafferana, but diversion of the lava flow saved the town. Initially the lava flow overtopped the barriers, so engineers used explosives near the source of the lava flow to divert it away from Zafferana.

Between 1995 and 2001 there was an increase in volcanic activity. In 2001 an eruption on the south side of Etna occurred during the tourist season and damaged the cable car station ([Figure 4.25](#)). In 2002–03, a large eruption destroyed the tourist station Piano Provenzana. Some of the footage of this eruption was used in the film *Star Wars: Episode III – Return of the Sith*.

Some eruptions continue for a very long time. One in 2008, for instance, lasted for 417 days. Eruptions in 2011 and 2012 led to the closure of Catania Airport several times. Starting in 2021, a series of explosive eruptions affected surrounding settlements, with rocks and volcanic ash falling as far away as Catania. Etna erupted 11 times in just three weeks.

The Global Volcanism Program reports that since 1955 Etna has experienced 17 VEI 1 eruptions, 24 VEI 2 eruptions and seven VEI 3 eruptions.



▲ **Figure 4.25** Damage to tourist infrastructure by lava flows at Etna, Sapienza Refuge ski area

There have been many responses to the threat of volcanic activity on Mount Etna. In the 1970s the government took photos of Etna temperature readings in order to monitor change. During an eruption in 1992 they built an earth barrier that was 200 m long and 20 m high to prevent lava from reaching Zafferana. It contained the lava flow for three months. US marines used explosives to blast a hole in the lava, and then dropped blocks of concrete into it to cool it down. During an eruption in 2002 evacuations occurred, schools were closed and a state of emergency was declared. The Institute of Vulcanology monitors radon gas, uses GPS to examine any changes in size or shape of the volcano, and uses very sensitive seismometers to detect earth tremors. Following the eruptions of 2002–3 the government pledged financial aid to farmers and those working in the tourism industry.

Despite the relatively low number of fatalities, Etna has proved disruptive to many local residents and to the tourism sector. Methods used to control the impacts of Mount Etna have included constructing earthen dams to divert lava, bombing the lava flows, cooling lava using sea water, blocking lava tunnels to reduce the lava flow and monitoring air quality (for sulphur dioxide) and ground deformation to make predictions about likely activity.

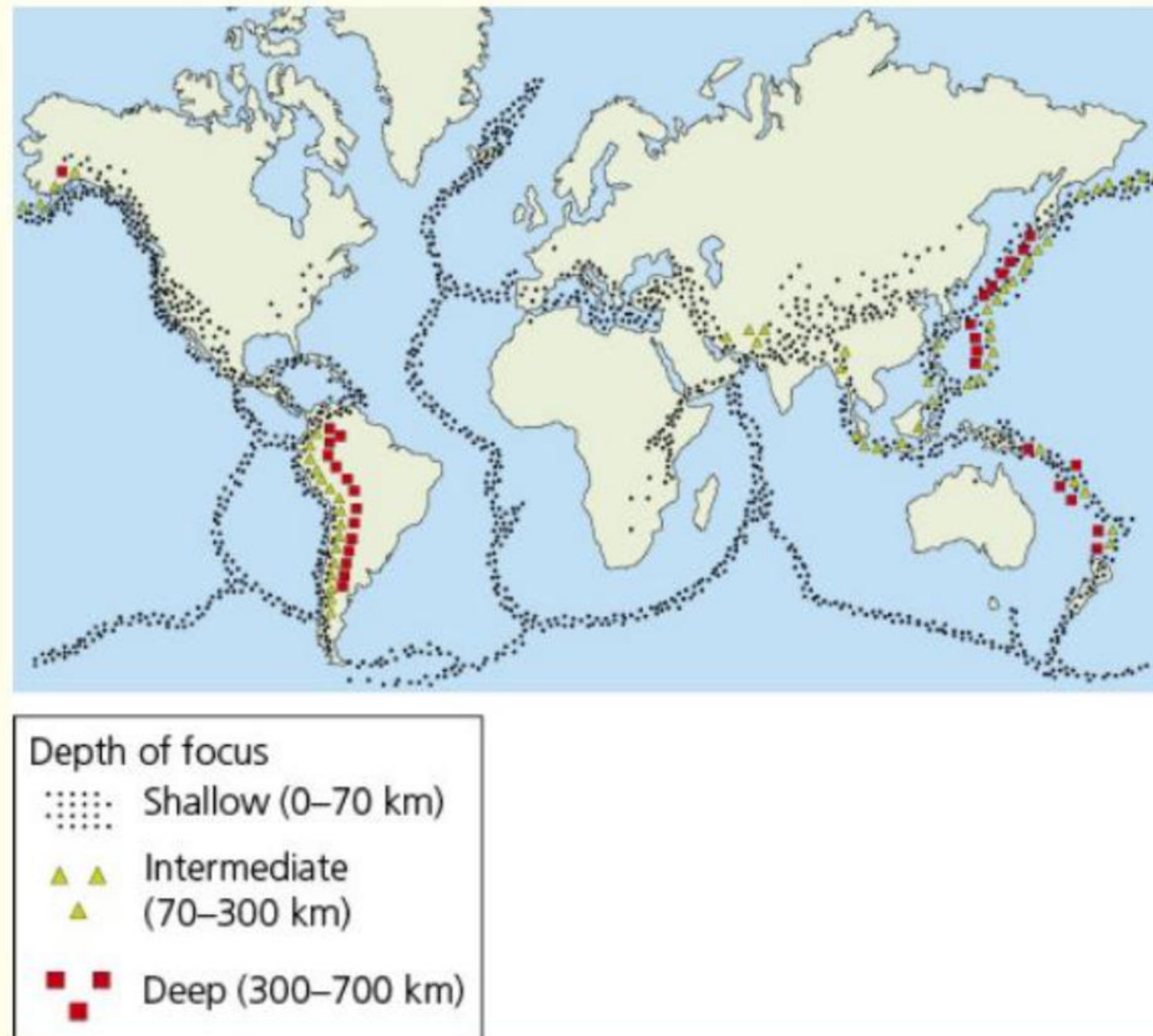
## Activities



- 1 Explain the impacts of volcanic activity on Mount Etna.
  - 2 Analyse ways in which the threats posed by Mount Etna have been managed.
- 

## Practice questions

- 1 [Figure 4.26](#) shows the distribution of shallow-, intermediate- and deep-focus earthquakes.



▲ **Figure 4.26** The distribution of shallow-, intermediate- and deep-focus earthquakes

- a Describe the distribution of shallow-focus and deep-focus earthquakes. [4]
  - b Suggest reasons to explain the distribution of shallow-focus and deep-focus earthquakes. [4]
- 2 [Figure 4.27](#) shows the summit of Mount Etna in Sicily.



▲ **Figure 4.27** The summit of Mount Etna

- a Identify **two** features of the volcano shown on the photograph. [2]
  - b Describe **two** hazards related to volcanoes. [4]
  - c Explain **two** reasons why people live in areas at risk from volcanic activity. [4]
- 3 Assess the extent to which the impacts of earthquakes can be managed. [7]